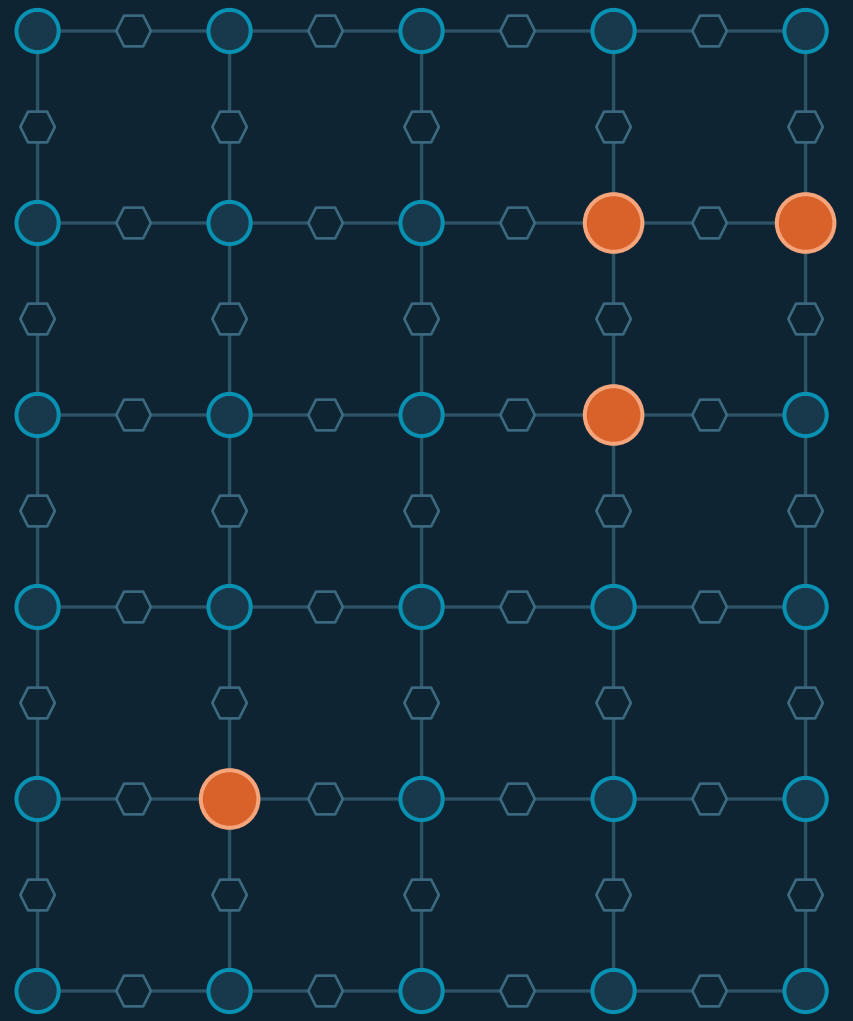
22AIE452 · DATA DRIVEN MATERIAL MODELING AND SIMULATION

MOFChat

Explainable, data-grounded metal-organic framework recommendation with validated LLM planning, multi-objective optimisation and surrogate models



TEAM 09 · FINAL REVIEW

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OVERVIEW

Outline

01

Problem and base paper

MOF selection, ChatMOF and its limitations

02

Literature and objectives

What prior work gives us, what is missing

03

Data and method

CoRE-MOF ETL, architecture, planner, validator, models

04

Results

Benchmark, model comparison, surrogates, explanations

05

Live application

The Streamlit app and demo questions

06

Lessons and conclusion

What we learned, limits, future work

Choosing a MOF means searching thousands of structures at once

Metal-organic frameworks are crystalline porous materials: metal nodes joined by organic linkers. Their tunable pores make them candidates for:

- **CO_2 capture**
- **H_2 and CH_4 storage**
- **Gas separation**
- **Catalysis**

A good choice needs database queries, several properties traded off together, and domain knowledge (which pore sizes admit which gases).

12,020

experimental MOFs in CoRE MOF 2019

65

different metals across the structures

27 %

have zero accessible surface area (pores too small for the N_2 probe)

ChatMOF made MOF search conversational, but unchecked

ChatMOF (Kang & Kim, Nature Communications 15, 4705, 2024)

A GPT-4 agent that plans tool calls: table search with LLM-written pandas code, MOFTransformer prediction, genetic-algorithm generation.

96.9 %	95.7 %	87.5 %
search	prediction	generation

Reported accuracy on its own question sets

Our question: keep this convenience, but check every answer against the data.

Limitations reported in the paper

1. **Token limits** 4 % of search tasks failed on large tables
2. **Logic errors** in the code the LLM writes
3. **Proprietary LLM** needs online GPT-4
4. **One model per property** a pretrained network for each target
5. **No validation** final answers are not checked against data

Prior work gives the parts; none checks LLM answers against data

Work	Reference	Idea	How MOFChat uses it
ChatMOF	Kang & Kim, Nat. Commun. 2024	LLM agent for MOF search	Base paper; same 12,020-MOF table
MOFTransformer	Kang et al., Nat. Mach. Intell. 2023	Pretrained MOF property model	We predict from descriptors, no 3D files
CoRE MOF 2019	Chung et al., J. Chem. Eng. Data 2019	Curated MOF structures + Zeo++	Our dataset; ETL verifies it
FT-Transformer	Gorishniy et al., NeurIPS 2021	Transformer for tabular data	Our embedding + regressor backbone
Trees vs deep learning	Grinsztajn et al., NeurIPS 2022	GBDT still strong on tables	Why we compare XGBoost vs FT
RAG / TableRAG / Chain-of-Table	Lewis 2020; Chen 2024; Wang 2024	LLMs over text and tables	SQL for numbers, RAG for knowledge
NSGA-II	Deb et al., IEEE TEVC 2002	Non-dominated sorting	Multi-objective MOF ranking
SHAP	Lundberg & Lee, NeurIPS 2017	Shapley explanations	Explains every prediction

All 15 references verified against publisher or arXiv records; full table in docs/LITERATURE_REVIEW.md

Objectives: exact, verifiable answers using free or local models

THE GAP

LLM agents make MOF search conversational, but they write code that can fail, depend on paid models and return numbers nobody checks.

We want the convenience of ChatMOF with the reliability of a database query.

1. **Exact answers**
   over all 12,020 MOFs, with free or local LLMs
2. **Safe LLM planning**
   validation, self-correction and fallback
3. **Multi-objective ranking**
   NSGA-II over several properties
4. **Predictions you can trust**
   uncertainty intervals and SHAP explanations
5. **Domain knowledge**
   retrieval over notes and the ChatMOF paper
6. **Measured, not claimed**
   labelled benchmark plus a held-out set

Review 1 failed: similarity search cannot satisfy number filters

REVIEW 1: TWO-TOWER RETRIEVAL

- FT-Transformer autoencoder indexed the MOFs
- Sentence-BERT embedded the question
- Same vector size, but never trained together, so retrieval ignored the question
- "Surface area > 4000" is a filter, not a similarity

9.4 %
dev set exact

5 %
held-out exact

FINAL: PLAN → VALIDATE → QUERY

- The LLM writes a structured JSON plan, never code
- A validator checks and repairs the plan
- Exact SQL filters; NSGA-II ranks
- FT-Transformer kept for similarity, where it fits

100 %
dev set exact

100 %
held-out exact

Lesson: constraint satisfaction over numbers is a database problem, not a similarity problem.

CoRE MOF 2019: 12,020 MOFs, checked before any modelling

Check	Finding	Action
Missing values, duplicate rows or names	none	-
Two column pairs identical	duplicates	dropped
Pore volume = void fraction / density	derived (error 6×10^{-6})	documented
Zero surface area (3,252 MOFs)	all have PLD < 3.31 Å	physics rule
Density header says cm^3/g	values are g/cm^3	label fixed
2,087 multi-metal MOFs	comma-separated metals	mof_metals table

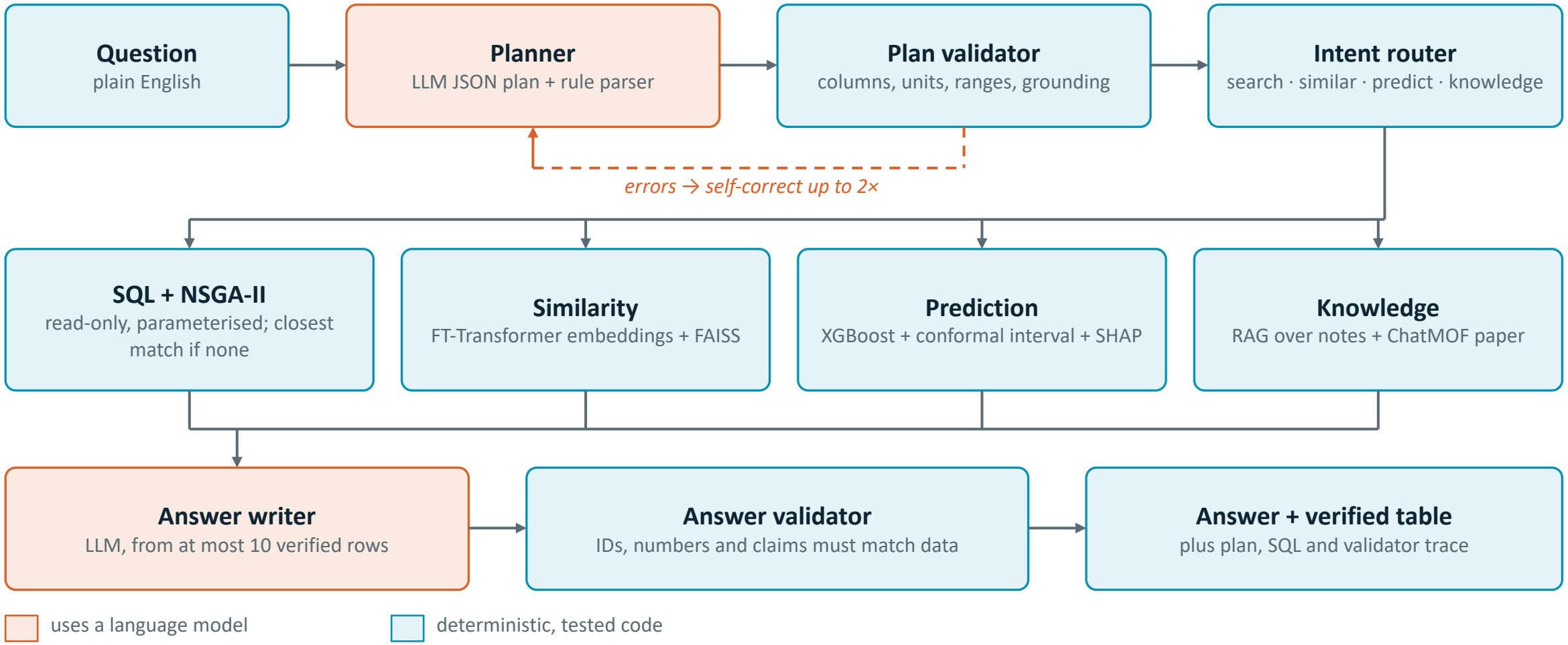


`data/coremof.db`

- **materials** – 15 columns, indexed
- **mof_metals** – one row per metal
- **columns_doc** – units and meaning

Zeo++ geometry, 3.31 Å N_2 probe. No gas-uptake labels in this table.

The LLM only plans and writes; everything else is checked code



The LLM writes a plan, not code, and the validator repairs it

“Which zinc MOFs have high surface area and low density?”

```
{ "intent": "search",
  "metals": ["Zn"],
  "objectives": [
    {"column": "Surface_Area", "direction": "max"},
    {"column": "Density", "direction": "min"} ],
  "limit": 5 }
```

Executor: whitelisted columns, bound parameters, read-only connection. Two objectives, so NSGA-II ranks the zinc MOFs.

```
SELECT * FROM materials WHERE MOF_ID IN
  (SELECT MOF_ID FROM mof_metals WHERE Metal IN (?))
```

Validator checks

- Column aliases resolved from one registry
- Units, operators and percentages normalised
- Impossible or invented values rejected
- Thresholds not in the question become preferences
- MOF IDs not in the question dropped
- Superlatives written as constraints repaired
- Application presets expanded and explained

Hybrid planner: facts the rule parser is certain of (numbers, metals, limits) are merged into the LLM plan.

14 free or local models behind one interface, with fallback

Provider	Models	Free limit
Google Gemini	3.1 Flash-Lite, 3.5 Flash, 3.8 Flash, Flash-Lite latest	~10–15 req/min
Groq	GPT-OSS 120B, GPT-OSS 20B, Qwen 3.8 27B	8,000 tokens/min
OpenRouter	Nemotron 3 Super / Ultra, Gemma 4 31B	~50 req/day
Ollama (offline)	Llama 3 8B, Llama 3.1 8B, Qwen 2.5 7B, Nemotron Mini	none, laptop CPU

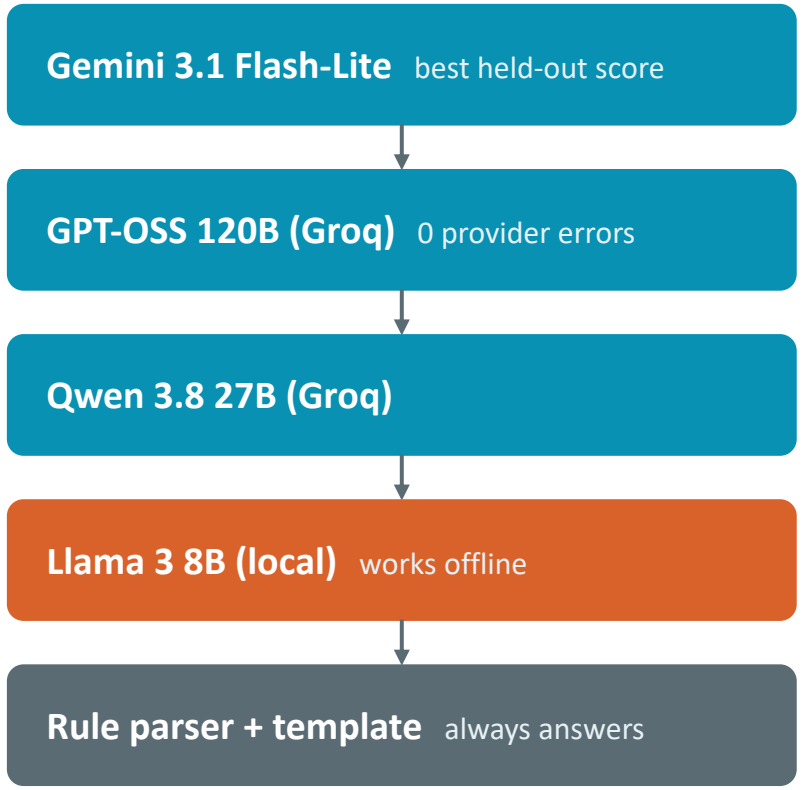
Each call is throttled to its free-tier budget. A rate limit or outage (429 / 503) moves the call to the next model.

14
models

4
providers

0
paid APIs

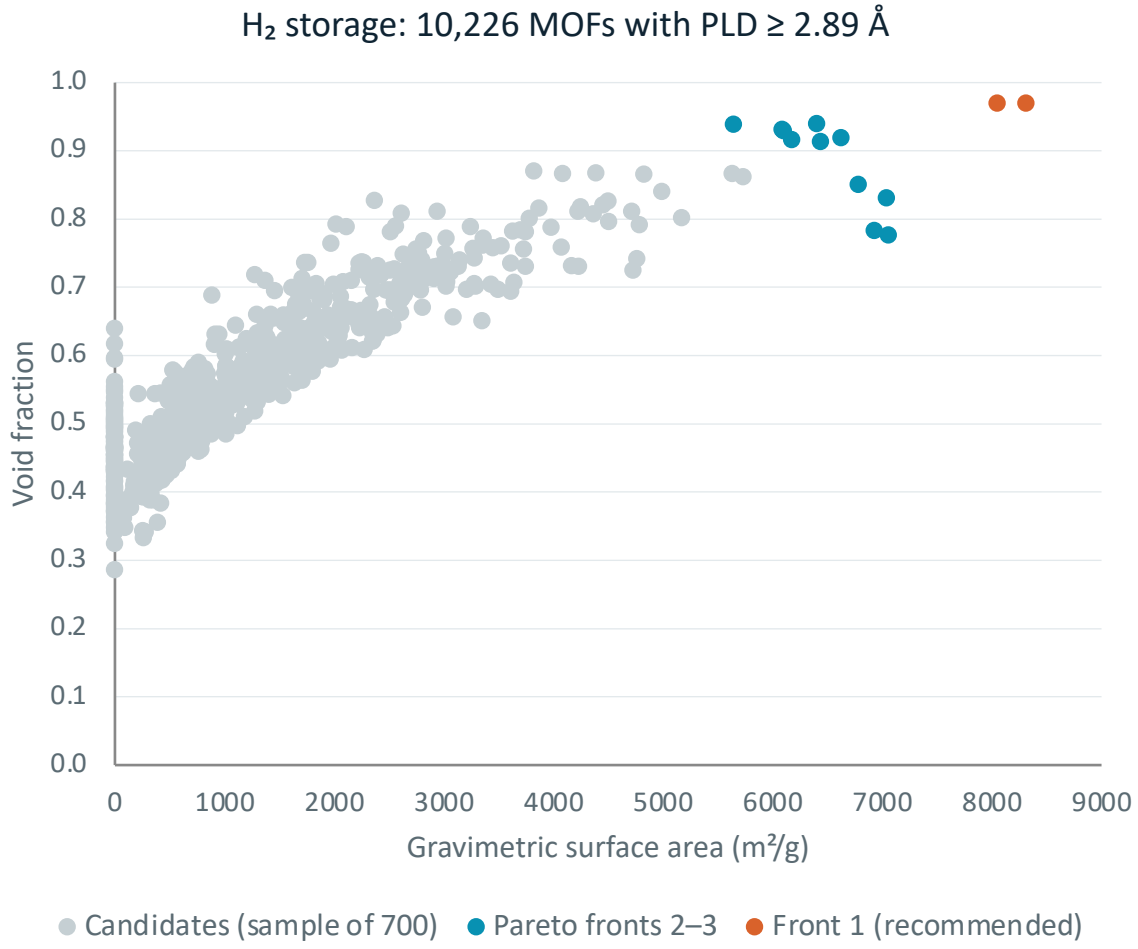
Default fallback chain



NSGA-II ranks candidates on several objectives at once

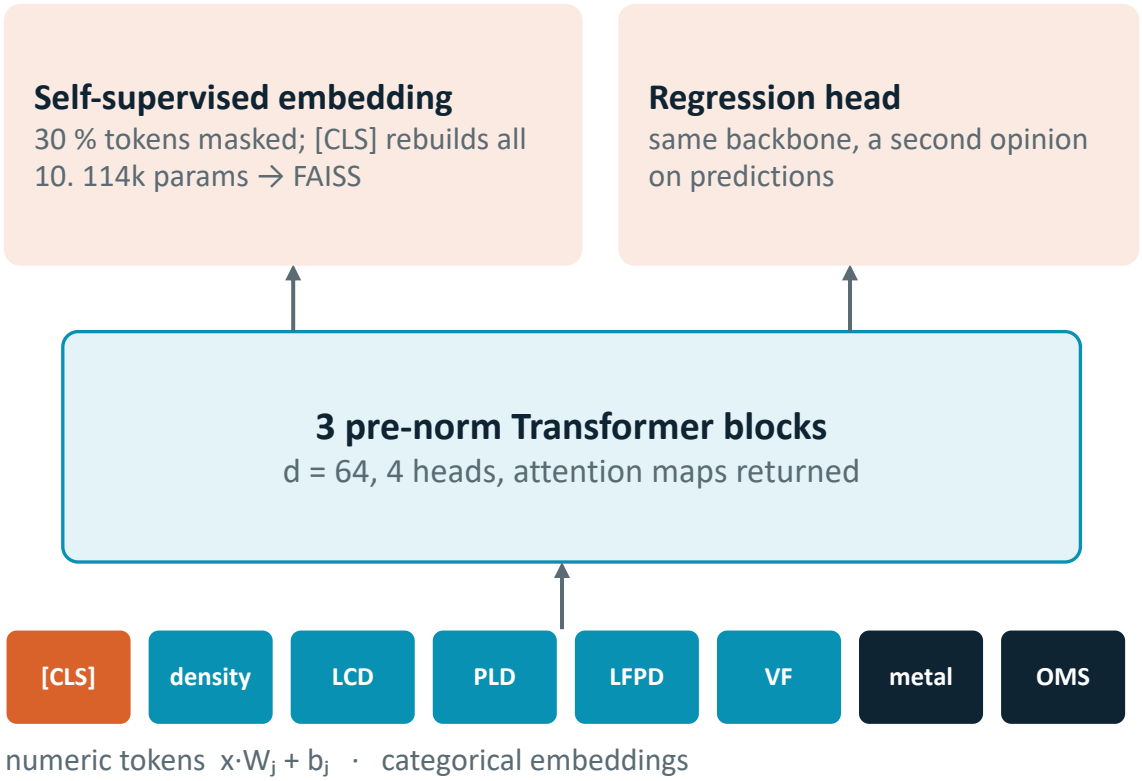
- **One objective** → plain sort
- **Two or more** → fast non-dominated sorting (exact O(n log n) sweep for two objectives) + crowding distance
- **Screening** → candidates in the worst 25 % of any objective removed first
- **No exact match** → closest MOFs by total constraint violation, and the user is told
- **Speed** → all 12,020 MOFs in 0.02 s (2 objectives), 2.6 s (3)

Preset	Constraints and objectives
H_2 storage	PLD ≥ 2.89 Å; max surface area, void fraction
CH_4 storage	PLD ≥ 3.80 Å; max volumetric area, void fraction
CO_2 capture	open metal site; PLD ≥ 3.30 Å; min LCD



Surrogates predict properties; an FT-Transformer powers similarity

FT-Transformer (Gorishniy et al., 2021)



Surrogate property models

- **Targets** accessible surface area; void fraction (pore volume excluded to avoid leakage)
- **Split** 70 / 10 / 10 / 10: train, early stopping, conformal calibration, test
- **Models** mean baseline, Ridge, Gaussian process, XGBoost (main), FT-Transformer
- **Physics rule** PLD below the 3.31 Å probe → surface area exactly 0
- **Trust** split-conformal 90 % intervals and TreeSHAP explanations

Retrieval adds knowledge; a validator keeps the LLM honest

Knowledge retrieval (RAG)

- 5 curated, referenced notes + the ChatMOF paper → 104 chunks
- all-MiniLM-L6-v2 embeddings, FAISS inner product
- Best match below 0.30 cosine → treated as off-topic
- Sources are shown with every answer

75 % hit@1

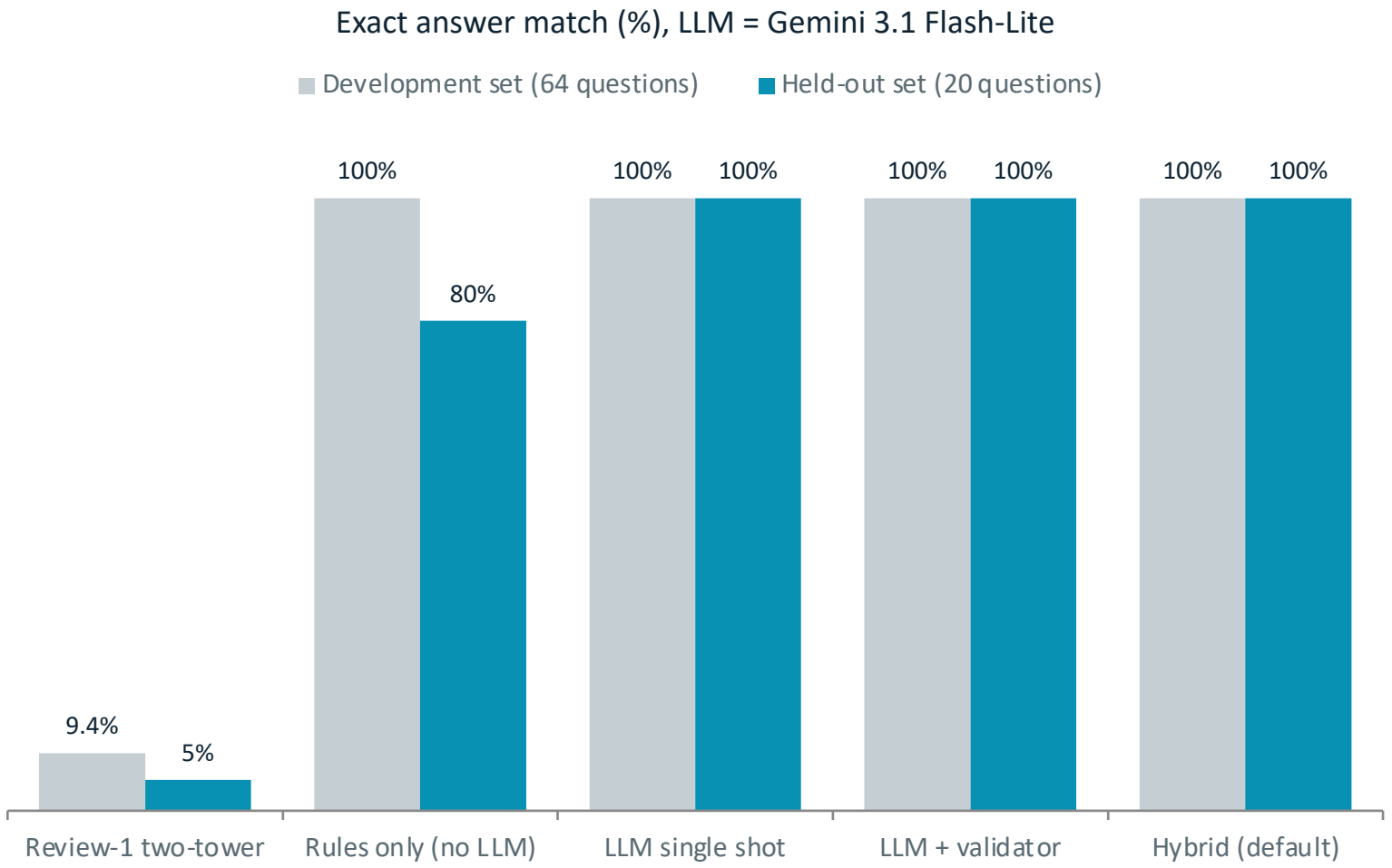
83 % hit@3

5 / 5 off-topic rejected

Answer validator

1. **Extract** every MOF ID and number, incl. “8 318”-style thousands
2. **Correct** IDs within edit distance 2 (ACOCLUS → ACOCUS)
3. **Trace** each number must match the data within rounding
4. **Check claims** “A, B, C have ≥ 6000 m^2/g”, “on the first Pareto front”
5. **Regenerate** once, listing the exact problems
6. **Flag** anything still unverified, above the verified table

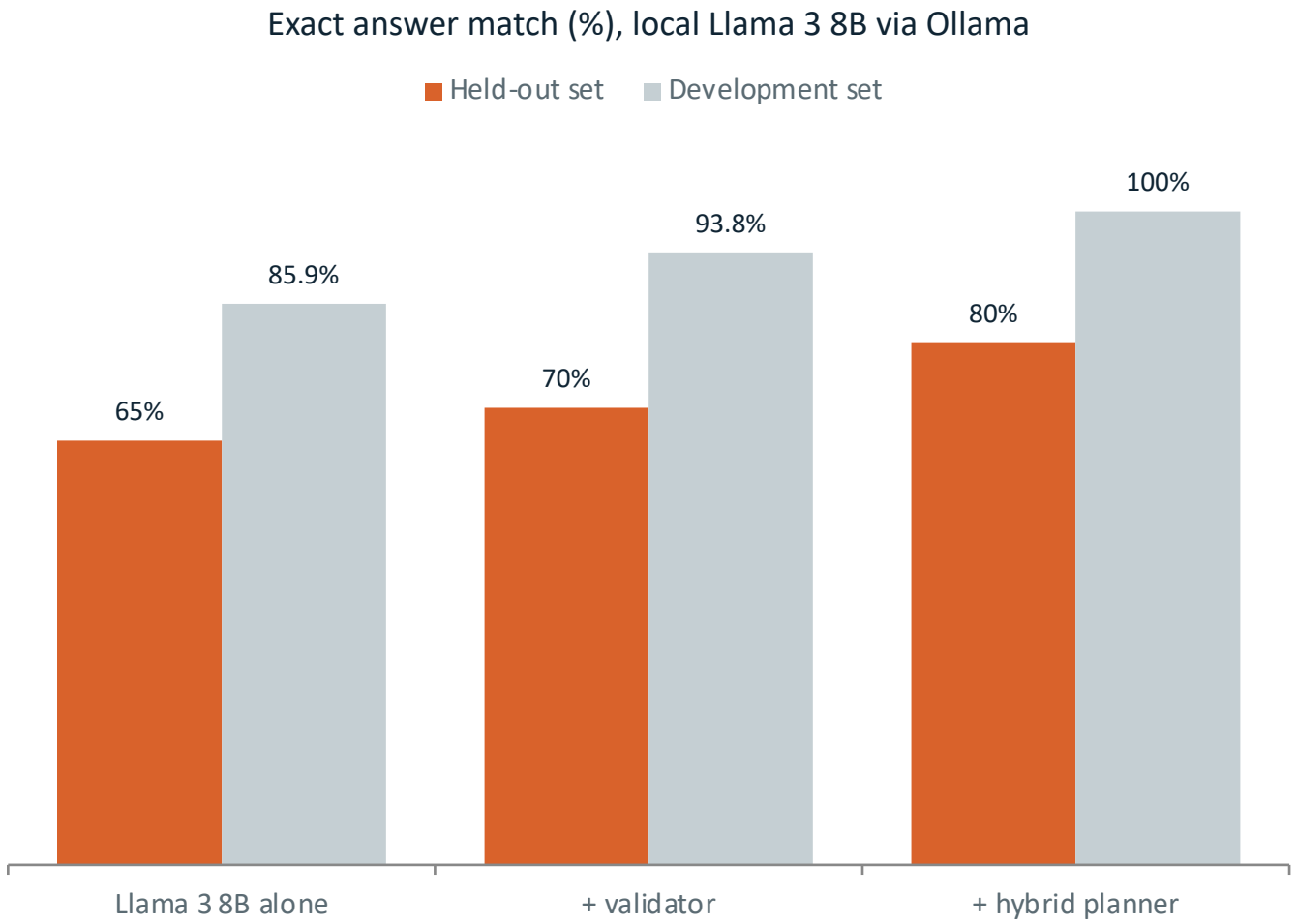
Exact answers jump from 5 % to 100 % on held-out questions



How to read it

- **Exact** = same MOFs (or count) as the hand-written gold plan
- **Dev 100 % is optimistic** for rules: they were tuned on it
- **Held-out** phrasings were written later, never tuned on
- **ChatMOF** reports 96.9 % on its own search set (different questions)

Validation and the hybrid planner make a small offline LLM usable



+15 pts

held-out accuracy added by the safety layers, fully offline

< 0.01 s

added by validation and the hybrid merge

With a strong cloud model the layers change nothing (100 → 100 → 100 %); they are insurance.

Remaining Llama errors: "natural gas" mapped to H_2, one extra objective, a count instead of a search.

Free cloud models tie on accuracy; reliability and speed differ

Model	Held-out: LLM alone	Held-out: hybrid	Dev: hybrid	Median call	Provider errors
Gemini 3.1 Flash-Lite (default)	**100 %**	**100 %**	**100 %**	**1.5 s**	**13 / 84**
GPT-OSS 120B (Groq)	95 %	95 %	100 %	2.4–4.7 s	0
Qwen 3.8 27B (Groq)	95 %	95 %	100 %	20 s	0
Nemotron 3 Super 120B (OpenRouter)	95 %	95 %	–	2.0 s	0
Llama 3 8B (local, offline)	65 %	80 %	100 %	1.5 s	0
Gemini 3.5 Flash	–	–	–	–	64 / 84 (quota)



Effectively tied
Three models miss only “density above 8 g/cm³”, answering with a count of 0.



Speed and reliability
Qwen 3.8 is a reasoning model and burns Groq's 8,000 tokens/min; Flash-Lite's errors were free-tier overload.



Not scored
Gemini 3.5 Flash ran out of daily quota, so its score would measure the fallback.

The answer validator catches invented numbers and false claims

Model	Drafts with a problem	Rewritten	Still flagged	Per answer
Gemini 3.1 Flash-Lite (default)	0 of 16	0	0	9.9 s
Llama 3 8B (local)	2 of 16	1	0	9.3 s
GPT-OSS 120B (Groq)	2 of 16	2	1	30.2 s
Qwen 3.8 27B (Groq)	0 of 16	0	0	26.4 s

16 full answers per model. Qwen 3.8: 7 answer calls hit Groq limits, so a template answer with the verified table was shown.

What it caught



Llama 3 8B

invented two densities and a “90 %”; garbled OFEUN06 → OFERUN06, corrected automatically



GPT-OSS 120B

claimed two MOFs were on “Pareto front 1”; NSGA-II ranked them on front 2 → shown with a warning

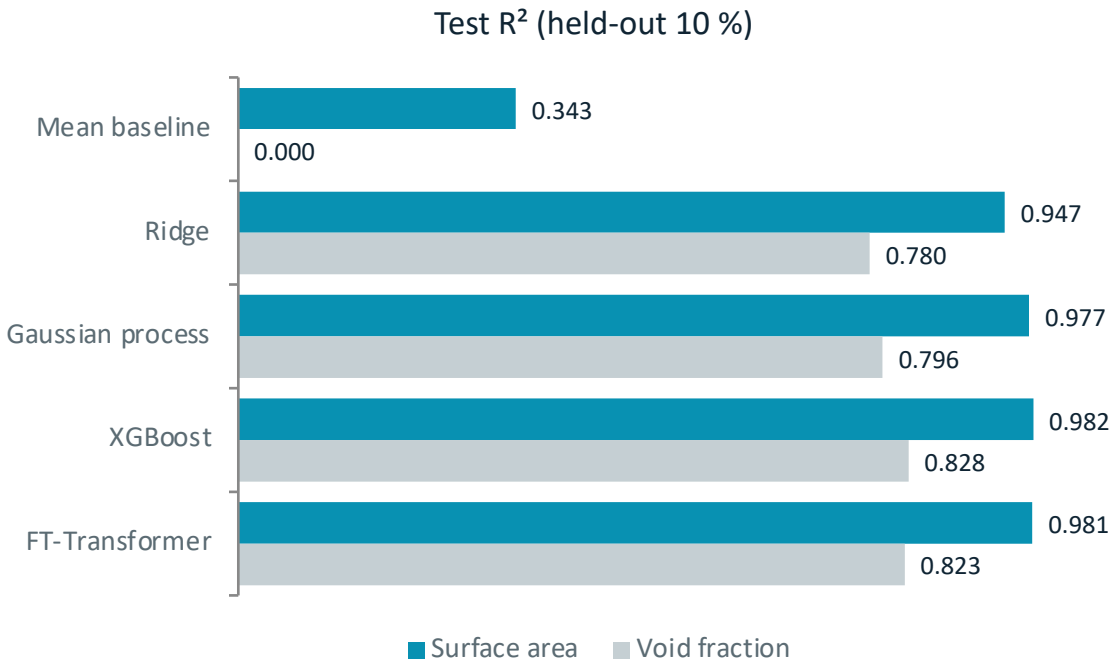
XGBoost: surface-area R^2 = 0.982 with calibrated intervals

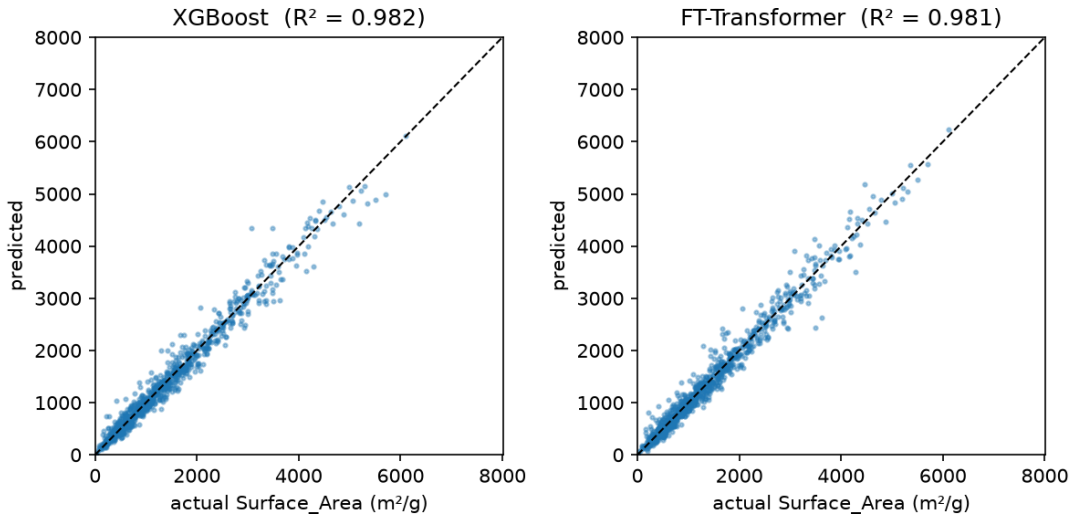
0.982
R^2 surface area (XGBoost)

90 m^2/g
mean absolute error

90.5 %
test coverage of the 90 % conformal interval

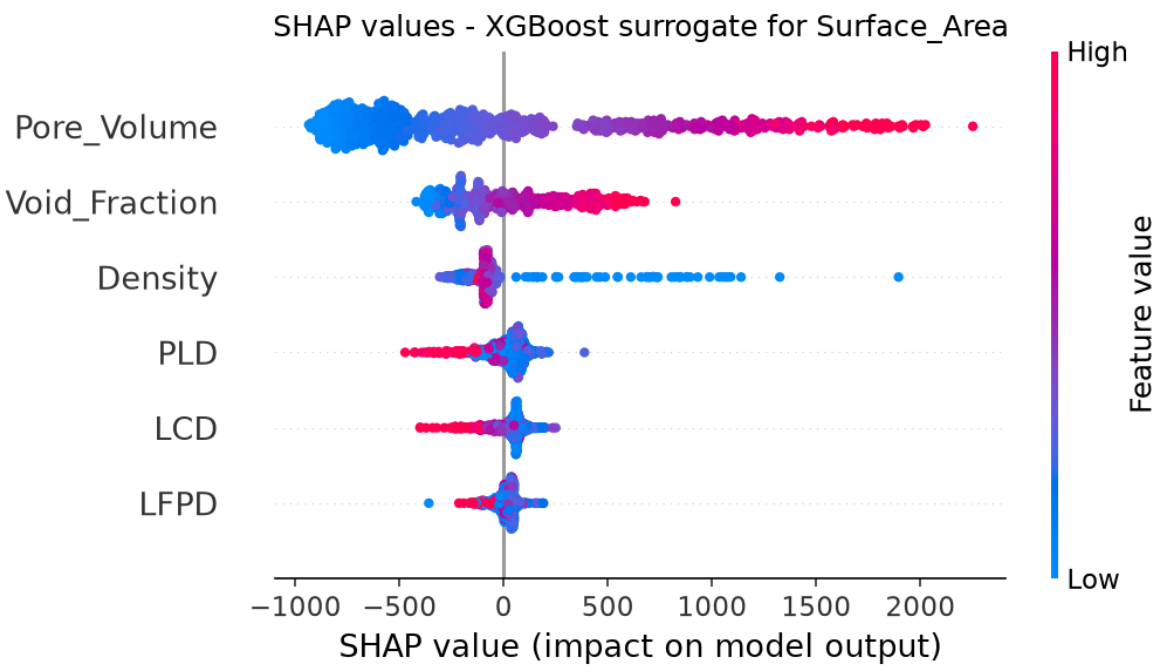
0.828
R^2 void fraction (XGBoost)





Predicted vs actual surface area on test MOFs. XGBoost trains 400× faster than the FT-Transformer (0.25 s vs 112 s) for the same accuracy.

SHAP drivers are physically sensible; embeddings find similar MOFs



Pore volume and void fraction dominate surface area; density dominates void fraction.

Similarity search: 1,000 queries × 10 neighbours

Neighbours from	Descriptor gap (σ)	Same metal
FT-Transformer embedding	0.077	**27.8 %**
k-NN on raw descriptors	0.068	22.8 %
Random MOFs	0.952	8.0 %

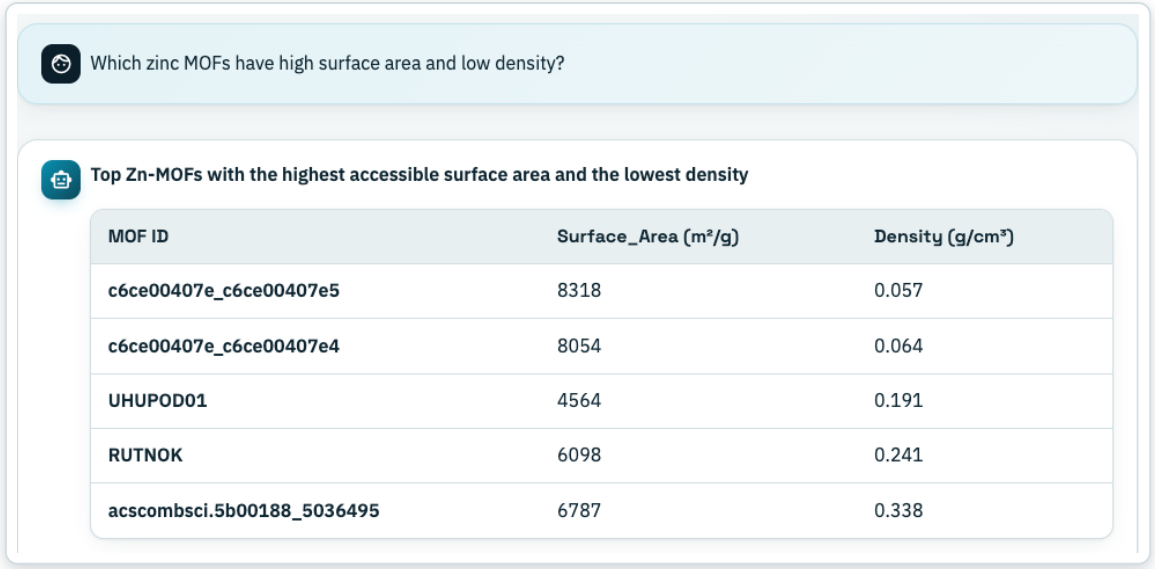
Nearly as close as k-NN on the numbers (k-NN is the upper bound on that metric), but more often the same metal, because metal and open-metal-site are input tokens. 62 % overlap with k-NN lists.

Seven of ChatMOF's nine limitations are fully addressed

ChatMOF limitation	What MOFChat does	Status
No intrinsic materials knowledge	RAG over curated notes + ChatMOF paper	**Done**
Token-context limits	SQL filters first; LLM sees ≤ 10 rows	**Done**
Logic errors in LLM code	JSON plan, validator, whitelisted SQL	**Done**
Restricted search space	all 12,020 MOFs searched and ranked	**Done**
Narrow topology coverage	all CoRE-MOF structures, 65 metals	**Done***
A pretrained model per property	surrogates trained on the table	**Done***
General, non-fine-tuned LLM	domain prompts, few-shot, presets	**Partly**
High compute cost	ms screening; XGBoost trains in 0.25 s	**Done**
Unvalidated output	answer validator, intervals, SHAP, benchmark	**Done**

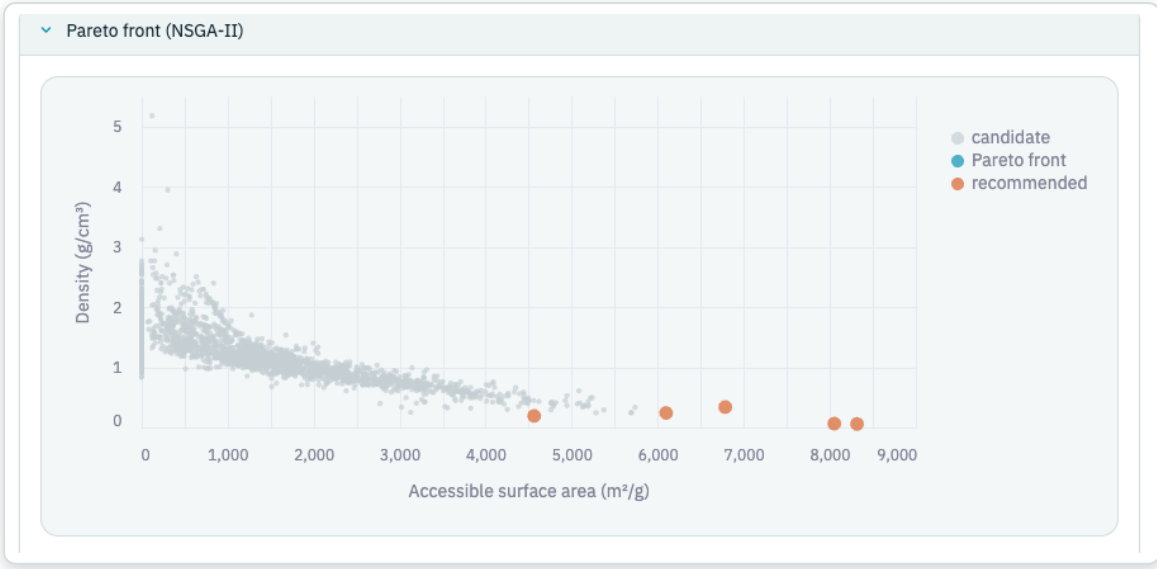
** for recommendation and for properties in the table: no structure generation, no gas uptake without uptake labels*

The app shows the answer, the verified table and how it was made



MOF ID	Surface_Area (m²/g)	Density (g/cm³)
c6ce00407e_c6ce00407e5	8318	0.057
c6ce00407e_c6ce00407e4	8054	0.064
UHUPOD01	4564	0.191
RUTNOK	6098	0.241
acscombsci.5b00188_5036495	6787	0.338

“Which zinc MOFs have high surface area and low density?”



The same answer's NSGA-II Pareto plot: recommended set in copper

Every answer also shows the plan, the SQL, the validator report and which model answered. With no language model at all, the rule planner and template answers still work.

Chat **Explore & optimise** **Predict & explain** **Evaluation**

Four lessons from building and testing MOFChat

Similarity ≠ constraints

A two-tower model needs paired training data, and numeric filters belong in a database. This is why review 1 scored 9.4 %.

LLMs and rules complement

Rules are exact but brittle on new phrasings; LLMs generalise but make schema errors. Validation and the hybrid merge combine both.

Use the app, not only the benchmark

The benchmark scores plans, not text. Using the app found four bugs (false ≥ 6000 claim, overridden presets, “90 %” false positive, “8 318”). Each now has a unit test.

Trees still win on small tables

On ~8,000 MOFs with 6 features XGBoost ties the FT-Transformer, trains 400× faster and gives exact SHAP.

What MOFChat does not do yet, and what comes next

	LIMITATION	NEXT STEP
1	**No gas-uptake labels**	→ Attach GCMC uptake data; reuse the surrogate pipeline
2	**No structure generation**	→ Recommend only; a generator (e.g. genetic algorithm) is future work
3	**Offline model at 80 %**	→ LoRA fine-tuning of Llama 3 on plan examples
4	**Free-text reasoning unchecked**	→ Validator covers IDs, numbers and claims; sources are shown
5	**Presets are heuristics**	→ Replace with learned rules once uptake data exists
6	**Descriptor-only similarity**	→ Add structure embeddings (e.g. MOFTransformer)

CONCLUSION

MOFChat turns the ChatMOF idea into a validated, measurable system that runs on free models or fully offline

100 %

held-out exact answers, default model (review 1: 5 %)

80 %

fully offline with Llama 3 8B, +15 pts from validation

0.982

surface-area R^2, 90.5 % interval coverage

14

free or local LLMs with automatic fallback

Exact answers over all 12,020 MOFs · NSGA-II multi-objective ranking · calibrated, explained predictions · literature grounding · an answer validator that keeps the LLM honest

Every number in this deck is produced by evaluate.py and recorded in docs/RESULTS.md.

References

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[2] Y. Kang, H. Park, B. Smit, J. Kim, “A multi-modal pre-training transformer for universal transfer learning in metal–organic frameworks,” Nat. Mach. Intell., 5, 309–318, 2023.

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[7] S.-A. Chen et al., “TableRAG: Million-token table understanding with language models,” NeurIPS, 2024.

[8] Z. Wang et al., “Chain-of-Table: Evolving tables in the reasoning chain for table understanding,” ICLR, 2024.

[9] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan, “A fast and elitist multiobjective genetic algorithm: NSGA-II,” IEEE Trans. Evol. Comput., 6, 182–197, 2002.

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[11] Z. Zheng, O. Zhang, C. Borgs, J. T. Chayes, O. M. Yaghi, “ChatGPT chemistry assistant for text mining and the prediction of MOF synthesis,” J. Am. Chem. Soc., 145, 18048–18062, 2023.

[12] T. M. Pruyn et al., “MOF-ChemUnity: Literature-informed large language models for metal–organic framework research,” J. Am. Chem. Soc., 2025.

Thank you

Questions and live demo

Code github.com/Sai-Mohith24/MOFChat-AI-Driven-Material-Recommendation-

Team 09 · 22AIE452 Data Driven Material Modeling and Simulation

